Ganesh Karayi

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Education

University of California, Berkeley — Berkeley, CA

Bachelor of Arts in Applied Mathematics and Data Science, GPA: 3.8/4.0 — *Expected May 2028*

- Relevant Coursework: Introduction to Artificial Intelligence, Principles and Techniques of Data Science, Probability and Random Processes, Data Structures and Algorithms; **In Progress**: Introduction to Machine Learning

Experience

IPick.AI — Cupertino, CA

AI Engineer — *Jan. 2026 – Present*

- Led 26 engineers across three ML workstreams building an XGBoost top-5 stock ranker and contextual-bandit stock-picking agents that learn from realized one-month returns
- Built a test-driven onboarding system (75 autograded tests, mocked APIs, per-ticket CI) that had new engineers shipping in their first sprint
- Led a team of 14 to ship a live equity knowledge graph of 4,300+ companies, building the LLM extraction pipeline (Llama 3.2, Claude) that pulled 198,000+ supplier and subsidiary mentions from SEC 10-K filings
- Cut false graph edges 93% (13,600 to 894) by replacing fuzzy substring entity matching with strict normalized-name resolution, removing links like "Honduras" resolving to a shoe brand

Shepherd Ventures — San Diego, CA

Research Fellow — *Oct. 2026 – Present*

- Caught look-ahead leakage that overstated worst-case protection by 2x (44.6% vs. a true 21.2%); the leaked data correlated 0.77 with event returns, nearly double the strongest legitimate signal (momentum, 0.41)
- Built a cached 21,504-backtest sweep engine whose parity check reproduced all 4,130 published runs exactly

Senior Quantitative Analyst — *Jan. 2026 – Oct. 2026*

- Created a fear-regime detection model with 100% recall on 8 labeled crashes over 20 years of history; separating noise alarms from real corrections raised precision from 0.21 to 0.57 and F1 from 0.34 to 0.73
- Diagnosed an XGBoost stress classifier (82.9% accuracy, 85.3% recall, 30.2% precision) through a false-positive/false-negative ledger across 10 stress periods, logging a rule fix for each
- Optimized K-Means-preselected portfolios on GARCH covariance for 54% downside capture vs. the S&P 100

LG NOVA — Remote

Software Engineer — *Aug. 2026 – Present*

- Root-caused silent data loss from a 500-row cap in a Flask/PostgreSQL endpoint; shipped keyset pagination
- Expanded semantic search coverage of PitchBook financial fields by 233% across Python and React/TypeScript repos, verified by a 38-test suite confirming gap-free pagination over 1,200+ rows
- Restored PowerPoint uploads to the AI document-analysis pipeline, adding PDF conversion as a fallback

DealPrint — Berkeley, CA

Data Science Research Intern — *Aug. 2025 – Jan. 2026*

- Designed a sale-price model from scratch: KNN comps weighted by NAICS hierarchy, geography, and deal type, feeding a regression on revenue, SDE/EBITDA margin, and state (0.25 MAPE on held-out deals)
- Scraped 50,000 M&A deals spanning 40 years via a custom Chrome extension; shipped it as a Streamlit dashboard

Projects

Solar Panel Market Quantile Analysis | *Python, statsmodels, NumPy, pandas* — 2026

- Implemented Minimum Distance Estimation (MDE) for quantile regression from scratch on 332,000 self-sourced records, fitting KDE sparsity and WLS for the asymptotic covariance; cross-validated against statsmodels

Activities

Vice President of Projects | *Open Project* — Aug. 2025 – Present

- Direct 40+ Project Managers across 15+ client projects, shipping 5 per semester, in a 300+ member organization

Skills and Interests

Languages: Python, SQL, Java, JavaScript
Libraries: PyTorch, scikit-learn, XGBoost, pandas, NumPy
Tools: PostgreSQL, Docker, AWS, Flask, Git
ML/AI: LLM Agents, NLP, Contextual Bandits, K-Means
Interests: Prediction Markets (Kalshi), Poker, Rock Climbing, Running